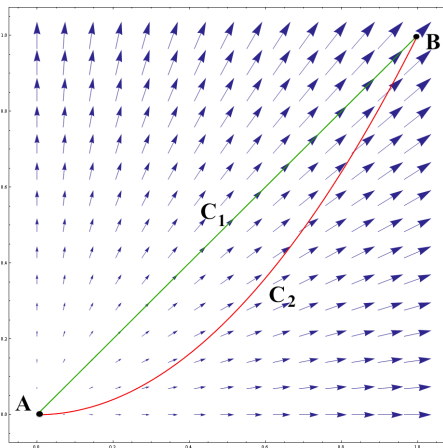
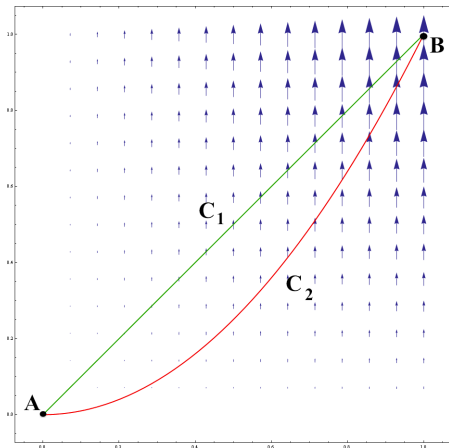


Topic: Path Independence

I. Introduction

Question: Does the amount of work required to move a mass from A to B in a force field depend on the path you take?

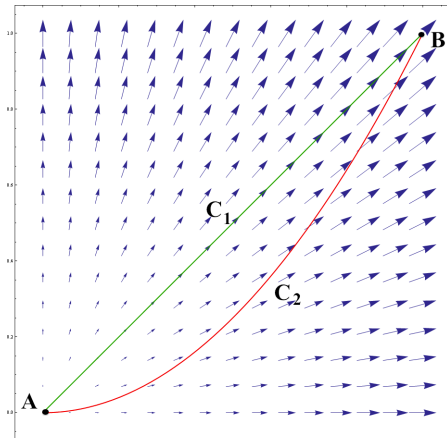


Answer: It depends on the force field.

II. Path Independence

Defn: Let C_1 be a curve from point A to B . The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is said to be **path independent** if for any other curve, C_2 , between A and B

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$$



III. Conservative Vector Fields and Potential Functions

Defn: A vector field $\mathbf{F}$ is **conservative** if $\int_C \mathbf{F} \cdot d\mathbf{r}$ is path independent.

Thm: If $\mathbf{F}$ is a conservative vector field then there exists a scalar function such that $\mathbf{F} = \nabla f$.

Defn: The f above is called a **potential for $\mathbf{F}$** .

Ex: Is $f(x, y, z) = x \sin(z^2) + e^{-y} + 13y$ a potential for $\mathbf{F}(x, y, z) = [\sin(z^2), -e^{-y} + 13, 2xz \cos(z^2)]$?
Is $\mathbf{F}(x, y, z)$ conservative?

Since $\nabla f = [\sin(z^2), -e^{-y} + 13, 2xz \cos(z^2)] = \mathbf{F} \implies f$ is a potential for $\mathbf{F}$

$\mathbf{F}$ is conservative since there exists an f such that $\nabla f = \mathbf{F}$

Ex: Is $f(x, y, z) = xy \ln(y) + z$ a potential for $\mathbf{F}(x, y, z) = \left[y \ln(y), \frac{x}{y}, 1 \right]$? Is $\mathbf{F}(x, y, z)$ conservative?

Since $\nabla f = [y \ln(y), x \ln(y) + x, 1] \neq \mathbf{F} \implies f$ is NOT a potential for $\mathbf{F}$

$\mathbf{F}$ might or might not be conservative, we need more tools to determine this.

Thm: Let C be a curve from A to B . If $\mathbf{F}$ is conservative (i.e. $\mathbf{F} = \nabla f$ for some f), then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(x, y, z) \Big|_A^B = f(B) - f(A)$$

Idea of Proof: Let $\mathbf{r}(t)$ be a parameterization of C with $t \in [a, b]$ then

$$\begin{aligned} \int_C \nabla f \cdot d\mathbf{r} &= \int_a^b \nabla f \cdot \mathbf{r}'(t) dt = \int_a^b \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] \cdot \left[\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right] dt \\ &= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\ &= \int_a^b \frac{df}{dt} dt \\ &= f(x(b), y(b), z(b)) - f(x(a), y(a), z(a)) \\ &= f(\mathbf{r}(b)) - f(\mathbf{r}(a)) \\ &= f(B) - f(A) \end{aligned}$$

There is a nice consequence of the previous theorem that is important in future discussions.

First a definition:

Defn: A curve, C , is **closed** if it has **no endpoints** and **encloses an area** (not necessarily **planar**). Any point on the curve can be thought of as an **initial** and an **end point of the curve**.

Thm: If $\mathbf{F}$ is **conservative** ($\mathbf{F} = \nabla f$) and C is closed then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 0$$

Idea of Proof: Take any point on C , let's call it A . Since C is closed, it could be thought of as a curve from point A to point A .

Then from the previous theorem, we have

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \nabla f \cdot d\mathbf{r} = f(x, y, z) \Big|_A^A = f(A) - f(A) = 0$$

Ex: The vector field $\mathbf{F} = [yz, xz, xy]$ is conservative and has potential $f(x, y, z) = xyz$ (i.e. $\mathbf{F} = \nabla f$). Find the work done by $\mathbf{F}$ on a mass along a curve, C , between $A(-1, 3, 9)$ and $B(1, 6, -4)$.

Soln: First note that since $\mathbf{F}$ is conservative, we have:

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \nabla f \, d\mathbf{r} = \int_C \nabla[xyz] \, d\mathbf{r} \\ &= xyz \Big|_{(-1, 3, 9)}^{(1, 6, -4)} \\ &= (1)(6)(-4) - (-1)(3)(9) \\ &= -24 + 27 = 3\end{aligned}$$

IV. Finding Potential Functions

If we are given a conservative vector field $\mathbf{F}$, then we know that there must be a corresponding potential function f such that $\nabla f = \mathbf{F}$.

But what is f ?

Let $\mathbf{F} = [M, N, P]$ (Note: M, N, P are all functions of x, y , and z .)

$$\text{Then } \nabla f = \mathbf{F} \implies \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] = [M, N, P] \implies \frac{\partial f}{\partial x} = M, \quad \frac{\partial f}{\partial y} = N, \quad \frac{\partial f}{\partial z} = P$$

So to find f , we must compute a sequence of **partial anti-derivatives**:

$$(i) \quad \frac{\partial f}{\partial x} = M$$

$$(ii) \quad \frac{\partial f}{\partial y} = N$$

$$(iii) \quad \frac{\partial f}{\partial z} = P$$

Ex: Let $\mathbf{F} = [e^x \cos y + yz, xz - e^x \sin y, xy + z]$

(a) Show $\mathbf{F}$ is conservative by finding an f such that $\mathbf{F} = \nabla f$

$$\mathbf{F} = \nabla f \implies [e^x \cos y + yz, xz - e^x \sin y, xy + z] = [f_x, f_y, f_z]$$

from above $\rightarrow$

$$f_x = e^x \cos y + yz$$

$$f = e^x \cos y + xyz + g(y, z) \quad \leftarrow \text{anti-der wrt } x$$

$$\text{from above } \rightarrow \quad xz - e^x \sin y = f_y = -e^x \sin y + xz + g_y \quad \leftarrow \text{partial wrt } y$$

$$g_y = 0 \quad \leftarrow \text{solve for } g_y$$

$$g = h(z) \quad \leftarrow \text{anti-der wrt } y$$

$$f = e^x \cos y + xyz + h(z) \quad \leftarrow \text{plug } g \text{ into } f$$

$$\text{from above } \rightarrow \quad xy + z = f_z = xy + h_z \quad \leftarrow \text{partial wrt } z$$

$$h_z = z \quad \leftarrow \text{solve for } h_z$$

$$h = \frac{z^2}{2} + C \quad \leftarrow \text{anti-der wrt } z$$

$$f = e^x \cos y + xyz + \frac{z^2}{2} + C \quad \leftarrow \text{plug } h \text{ into } f$$

$$f = e^x \cos y + xyz + \frac{z^2}{2} \quad \leftarrow \text{set } C = 0$$

(b) Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is a curve from $A : (0, 0, 2)$ to $B : (2, \pi/2, 0)$

Knowing f such that $\mathbf{F} = \nabla f$ makes this much easier since we don't need a parameterization for C . We don't even need to know what C is.

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_A^B \nabla f \, d\mathbf{r} = f(B) - f(A)$$

and computing

$$f(B) = e^2 \cos(\pi/2) + (2)(\pi/2)(0) + \frac{(0)^2}{2} = 0$$

$$f(A) = e^0 \cos(0) + (0)(0)(2) + \frac{(2)^2}{2} = 1 + 2 = 3$$

which gives

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 0 - 3 = -3$$

Ex: Let $\mathbf{F} = [y, x, 4]$.

(a) Show $\mathbf{F}$ is conservative by finding an f such that $\mathbf{F} = \nabla f$

$$\mathbf{F} = \nabla f \quad \implies \quad [y, x, 4] = [f_x, f_y, f_z]$$

from above $\rightarrow$	$f_x = y$	
	$f = xy + g(y, z)$	$\leftarrow$ anti-der wrt x
from above $\rightarrow$	$x = f_y = x + g_y$	$\leftarrow$ partial wrt y
	$g_y = 0$	$\leftarrow$ solve for g_y
	$g = h(z)$	$\leftarrow$ anti-der wrt y
	$f = xy + h(z)$	$\leftarrow$ plug g into f
from above $\rightarrow$	$4 = f_z = 0 + h_z$	$\leftarrow$ partial wrt z
	$h_z = 4$	$\leftarrow$ solve for h_z
	$h = 4z + C$	$\leftarrow$ anti-der wrt z
	$f = xy + 4z + C$	$\leftarrow$ plug h into f
	$f = xy + 4z$	$\leftarrow$ set $C = 0$

(b) Compute $\int_{(1,1,1)}^{(2,3,-1)} y \, dx + x \, dy + 4 \, dz \quad \left(= \int_{(1,1,1)}^{(2,3,-1)} \mathbf{F} \cdot d\mathbf{r} \quad \text{where} \quad \mathbf{F} = [y, x, 4] \text{ \& } d\mathbf{r} = [dx, dy, dz] \right)$

From part (a), we found that $f(x, y, z) = xy + 4z$

and therefore

$$\begin{aligned} \int_{(1,1,1)}^{(2,3,-1)} \mathbf{F} \cdot d\mathbf{r} &= \int_{(1,1,1)}^{(2,3,-1)} \nabla f \cdot d\mathbf{r} = f(2, 3, -1) - f(1, 1, 1) \\ &= [(2)(3) + 4(-1)] - [(1)(1) + 4(1)] \\ &= 2 - 5 = -3 \end{aligned}$$

V. Conservative Test for Vector Fields

In general, to show a vector field is conservative by finding f such that $\mathbf{F} = \nabla f$ can be time consuming and challenging.

Fortunately, there is a test to determine whether a vector field, $\mathbf{F}$, is conservative.

Conservative vector field test: Let $\mathbf{F} = [M, N, P]$ be a vector field.

Assuming the partial derivatives of M, N, P exist, then $\mathbf{F}$ is conservative if

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y} \quad \text{or} \quad P_y = N_z, \quad M_z = P_x, \quad N_x = M_y$$

This is also equivalent to showing

$$\text{curl}(\mathbf{F}) = \mathbf{0} \quad \text{or} \quad \nabla \times \mathbf{F} = \mathbf{0}$$

Ex: Is $\mathbf{F} = [2x - 3, -z, \cos z]$ conservative?

Since $P_y = (\cos z)_y = 0$ and $N_z = (-z)_z = -1 \implies P_y \neq N_z$

So $\mathbf{F}$ is not conservative.

Ex: Let $\mathbf{F} = [2xz, 3z, x^2 + 3y]$. Is $\mathbf{F}$ is conservative?

Check:

$$P_y = (x^2 + 3y)_y = 3 \quad \text{and} \quad N_z = (3z)_z = 3 \quad \implies \quad P_y = N_z \quad \checkmark$$

$$M_z = (2xz)_z = 2x \quad \text{and} \quad P_x = (x^2 + 3y)_x = 2x \quad \implies \quad M_z = P_x \quad \checkmark$$

$$N_x = (3z)_x = 0 \quad \text{and} \quad M_y = (2xz)_y = 0 \quad \implies \quad N_x = M_y \quad \checkmark$$

Therefore, $\mathbf{F}$ is conservative

Ex: Let $f = x^2z + 3yz$ and $\mathbf{F} = 2xz\mathbf{i} + 3z\mathbf{j} + (x^2 + 3y)\mathbf{k}$.

(a) Is f the potential for $\mathbf{F}$?

Check:

$$\nabla f = [(x^2z + 3yz)_x, (x^2z + 3yz)_y, (x^2z + 3yz)_z] = [2xz, 3z, x^2 + 3y] = \mathbf{F} \checkmark$$

Yes, f is the potential for $\mathbf{F}$

(b) Suppose you know the work required to move a mass through $\mathbf{F}$ along a curve between the points $(1, 1, 1)$ and $(d, 2d, 2)$ is $10 \text{ N} \cdot \text{m}$.

What are the possible terminal points, $(d, 2d, 2)$, for C ?

Since $\mathbf{F}$ is conservative, we have

$$\begin{aligned} 10 &= \int_C \mathbf{F} \cdot d\mathbf{r} = \int_{(d, 2d, 2)}^{(1, 1, 1)} \nabla f \cdot d\mathbf{r} = f(d, 2d, 2) - f(1, 1, 1) \\ &= [(d)^2(2) + 3(2d)(2)] - [(1)^2(1) + 3(1)(1)] \\ &= 2d^2 + 12d - 4 \end{aligned}$$

$$2d^2 + 12d - 4 = 10$$

$$2d^2 + 12d - 14 = 0$$

$$2(d^2 + 6d - 7) = 0$$

$$2(d - 1)(d + 7) = 0 \implies d = 1, -7 \implies \text{Possible terminal points: } (1, 2, 2) \text{ or } (-7, -14, 2)$$